

The Mesh Programme: Formula Compendium

A single-place reference for the load-bearing relations of the rope corpus, organised by sector. Every entry carries its registry provenance and its status. Compiled at v3.17.0 (577 registered claims, 539 code-backed). The registry (claims.yaml) is the authority; where this document and the registry disagree, the registry wins.

How to read this document

STATUS grades, as used throughout the corpus: Derived — proved from stated premises, all of which are registered. Modeled — computed within a model whose premises are stated but not all derived. EFT-constrained — fixed by effective-theory matching. Conjecture — asserted with its conditionality on the face. Open — registered as unresolved. Failed-and-kept — refuted, retained on display.

Symbols. T (or T_0) strand tension [N]; μ line density [kg/m]; Σ stiffness / tension energy density [J/m³]; a mesh spacing [m]; w strand width; κ the locking modulus [J/m²] (NOT the surface gravity κ of the gravity sector, nor ELEC-021's Coulomb coefficient — a registered name collision, rename queued); K the coarse-grained XY stiffness [J]; J the per-link locking energy [J]; l_q the winding's source length; $g = l_q/a$; d_c strand thickness; c speed of light.

Conditionality. Everything vacuum-facing (Sigma_vac, the mesh pair (a, T0), and every number derived from them) is conditional on FND-037 at the FND-040 floors. Absolute scale is irreducible by the corpus's own theorem (FND-MATTER-005): one calibration has been spent (the electron mass), and it is spent once.

1. FOUNDATIONS — the medium's primitives

The framework starts from strands with tension, a mesh spacing, and a locking interaction at crossings. Everything else is built from these.

J = T^2 / kappa form)	locking energy per endpoint (exact XY)
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FND-001, Derived. Endpoint force balance. [Under the PRED-003-ETA enslavement this sharpens to $J = T a/2$.]

$$K = J / a \quad (\text{scalar}); \quad K = 2J/a \quad \text{coarse-grained stiffness (director form)}$$

FND-002, Derived. Corrects an earlier 3J/a. Load-bearing for the coupling sector.

$\Pi = \kappa a / T$ the single dimensionless coupling

FND-005, Derived. The parameter count: $\{T, \kappa, a\}$ carry exactly one dimensionless combination.

$$\sigma = 3 T_0 / a^2 \quad \Leftrightarrow \quad T_0 = \sigma a^2 / 3$$

FND-017 / GRV-076, Derived invariance. The relation that lets the stiffness and the mesh pair be solved jointly.

$$E_{\text{vortex}} = \pi K \ln(R/a_{\text{eff}}) + E_{\text{core}}, \quad a_{\text{eff}} = 0.18 a, \quad E_{\text{core}} = 5.448 K$$

FND-008/009 + benchmarks/micromech/defect_cores.py, Derived. The core cutoff is computed, not assumed.

$$E_{\text{pair}} = 2 \pi K \ln(d/a) \quad \text{vortex-antivortex (2D Coulomb, BKT)}$$

FND-009, Derived.

$$S_{\text{eff}} = (K/2) \int (\text{grad } \theta)^2 \quad \text{leading-order effective action}$$

EFT-001, EFT-constrained; Gamma-convergence of the discrete locking energy on vortex-free fields (FND-003).

$$u(x) = T_0 \left(x/2 - x^2/8 + \dots \right), \quad x = (\text{grad } g)^2$$

FND-040, Derived. The arc-length expansion of an inextensible constant-tension strand — the NEGATIVE quartic is the medium's softening.

$$\mu = T / c^2 \quad \text{line energy equals tension}$$

HBAR-001 / GRV-077, Derived identity; with $c = \sqrt{T/\mu}$ it is a restatement, and is used as such.

2. ELECTROMAGNETISM — charge, fields, and the fine-structure chain

$$q = Lk \quad \text{charge = topological linking number} \\ (\text{integer})$$

EM-001 / GG-006, Derived. Charge is a conserved integer; the Coulomb field is rope-count flux geometry. The screw/torsion Goldstone carries CHARGE (winding); the separate collective transverse pair carries LIGHT (EM-RECON-025).

$$\text{Maxwell} \quad \text{from Bianchi identity + Chern-Weil in } d = 3$$

EM-003, Derived. The field equations are geometric identities of the bundle, not postulates.

$$Z_0 = \sqrt{\mu_0/\epsilon_0} = 376.73 \text{ ohm}; \quad c^2 = 1/(\mu_0 \epsilon_0)$$

EM-002, Derived (structural constants). $\mu_0 = 1/(\epsilon_0 c^2)$ forced by the carrier wave equation via Faraday (EM-RECON-028).

$$Z_{\text{medium}} = \sqrt{T \mu} = T / c \quad \text{intrinsic rope impedance}$$

OPT-006, Derived. Yields $\epsilon_{\text{med}} = 1/T$ — the medium's permittivity is the inverse tension (PRED-003-DICT).

$$\alpha = l_q^2 T / (4 \pi \hbar c) \quad \text{the impedance reduction}$$

PRED-003-DICT, Modeled. l_q is the winding's SOURCE length; equivalently $\hbar = T l_q^2 / (4 \pi \alpha c)$.

$$\kappa_{\text{lock}} = 2T/(\eta a), \quad \eta = 1 \Rightarrow \kappa = 2T/a, \quad l_{\text{lock}} = a/2, \quad J = T a/2$$

PRED-003-LOCK / PRED-003-ETA, Modeled (conditional on P1 one-metric applicability, P2 single-symbol-T). $\eta = 1$ by one-metric uniqueness: one wave operator cannot carry two stiffnesses.

$$\alpha = g^2 T a^2 / (4 \pi \hbar c), \quad g = l_q/a; \quad \lambda = g^2 / (4 \pi)$$

PRED-003-ETA, Modeled. g is the corpus's single remaining mesoscopic unknown (FND-044).

$$g = l_q/a = 43.0 \cdot \kappa_{\text{pack}}^{(1/6)} \quad [82.6 \text{ at the 5\% floor, } 108.0 \text{ continuum}]$$

FND-041, Modeled. The sixth-root law: $a \sim \kappa^{(-1/3)}$ (FND-038/040) and $l_q \sim T^{(-1/2)}$ (R1).

$$\kappa_0 = c / \sqrt{\epsilon_0 \Sigma} \quad \text{circulation quantum normalization}$$

EM-RECON-027/028, Modeled; the value is Σ -normalization dependent (Σ_{eff} -normalized after FND-038).

$$1/\alpha = 4 \pi^3 D_E = 137.0605 \quad (+178.8 \text{ ppm}), \quad D_E = 1.1051029 \text{ blind}$$

ALPHA-DE-CHAIN, Modeled. A distinct route: no continuous dials, one discrete input, three named external gates.

$$\sin^2 \theta_W = 1/(3 \sqrt{2}) = 0.2357$$

EW-001, Modeled. From winding/Hopf geometry.

$\Delta n(\text{par}) : \Delta n(\text{perp}) = 3 : 1$, sign NEGATIVE (vacuum birefringence)

EM-RECON-016, Derived given the field mapping. A double qualitative discriminator against QED's 7:4 positive; PVLAS excluded the ATLAS identification.

3. LIGHT AND CLASSICAL OPTICS — the transverse channel

WHAT THE CARRIER IS, stated carefully, because the corpus corrected itself here. Light is the COLLECTIVE TRANSVERSE displacement mode of the mesh — the transverse Goldstone pair of broken translation symmetry, two polarization states (s_x, s_y), gapless at all crossing strengths (EM-RECON-025, Modeled). It is NOT the torsion/screw mode. GRV-020's Corollary 1 allocated "torsion dynamics = light" by topology, and that allocation OVER-REACHED: the single screw field carries ONE state against the photon's two and was killed on polarization by three computed identities — Malus washout, the Beth angular-momentum sign, and birefringence mode-count (EM-RECON-022/023, kill class 4). The tension was split into its own claim (GRV-102) and resolved in structure by Commission K. The screw mode retains its OTHER job: its winding is charge ($\pi_1(S^1) = \mathbb{Z}$, GG-006 — that half of Corollary 1 is untouched and load-bearing).

Live conditions on the carrier, on its face: the bending falsifier $B \leq T_0 a^2/12$, the mandatory three-pin fork, and the dispersive-spacing cost ($a_{\text{disp}} \leq 9.3e-28$ m against a coverage scale of $6.0e-17$ m). The optics results below ride on the transverse channel and were always written on it; nine of ten OPT claims are Derived, and they are unaffected by the carrier correction.

$\omega(k) = c k$ non-dispersive propagation

EM-005, Derived. Phase velocity independent of k .

$c = \sqrt{T/\mu}$ wave speed from the medium

Standard string result on the corpus's primitives; verified pointwise in curved backgrounds to ~2% (GRV-048).

wavefront = envelope of secondary wavelets, advancing at c (Huygens)

OPT-001, Derived.

$I(\theta) \propto \text{sinc}^2(\pi a \sin(\theta) / \lambda)$ single slit

OPT-002, Derived.

$I \propto 1 + \cos(2\pi \Delta / \lambda)$ two-slit interference

EM-004, Derived, from linear superposition alone.

$r = (Z_1 - Z_2)/(Z_1 + Z_2); \quad t = 2 Z_1/(Z_1 + Z_2)$ Fresnel amplitudes

OPT-008, Derived, from the impedance mismatch.

$\theta_c = \arcsin(n_2/n_1)$ total internal reflection

OPT-005, Derived.

$\tan \theta_B = n_2/n_1$ Brewster angle

OPT-009, Derived.

$\omega_n = n \pi c / L; \quad f_1 = c/(2L); \quad \omega_{\text{cutoff}} = \pi c / a$

OPT-003 / OPT-004, Derived. The mesh cutoff $k \leq 1/a$ is what makes cell-scale loops mode-free (FND-MATTER-053).

4. GRAVITY — induced metric and Einstein-Hilbert dynamics

Gravity is not postulated: it is the effective metric seen by transverse excitations, with its dynamics induced Sakharov-style from the weave band.

Zero-point theorem: a uniform network of ANY tension density is FLAT.

GRV-004, Derived. Rest tension does not gravitate; only deviations do. Structural — independent of Sigma's value.

$\mu u_{tt} = d_a (T_a d_a u)$ the gapless transverse wave operator

GRV-029/055, Derived. Four coefficient functions (μ, T_x, T_y, T_z).

$\mu = B/\alpha, \quad T_a = \alpha B/b_a^2; \quad \alpha B = \sqrt{\mu \square T}$

isotropic: $b = (T \mu)^{1/4}, \quad \alpha = T^{3/4} \mu^{-1/4}$

GRV-029, Derived. An exact bijection on the positive cone: the photon sector is ONE-METRIC because there is no fifth function for a second metric to live in.

$S_{\text{induced}} \rightarrow$ Einstein-Hilbert (m-odd, IR-universal remainder; parameter-free tensor pattern)

GRV-025, Derived. The absorption exam: measured tensor ratios match the EH pattern.

$G = c^3 a^2 / (16 \pi \zeta \hbar), \quad \zeta = 1.208$

GRV-075/095, Modeled/Derived-conditional. The Sakharov route; a selected at the Planck class ($a = 0.80 l_P$ after the GRV-101 correction).

1.751" deflection, 43.0"/century Mercury, Shapiro delay, Nordtvedt — unconditional within the corpus

GRV-029 consequence, Derived. $\gamma = 1$ with the scalar screened at lattice range (GRV-028).

$S = A/4$ from $T_H = \kappa/2\pi$ with $dM = T dS$

GRV-039, Derived (numerically verified). Horizon thermodynamics from the ledger mechanism.

$T_{\text{inf}} = (\beta K \hbar / m^*) \kappa$ the Hawking law-form

GRV-087, Modeled. σ cancels exactly — the third cancellation.

$A^* = (\beta/0.23) \Sigma a^3 \hbar / (\chi c); \quad n_q = A^*/\hbar = 1.1e-4.4.6e-4$

GRV-092, Modeled. The snap action is depth- and mass-blind: the horizon secretes a universal action constant. Sub-quantum — the whisper is classical radiation at leading order.

$\hbar = T_0 \cdot [l_q^2/(4 \pi \alpha)] / c$ and $A^* = T_0 \cdot [(3 \beta/0.23 \chi)) a \hbar] / c$

GRV-093, Modeled. ONE FORM — tension $\times$ area / c — across two sectors; the $\hbar$ gap is the ratio of two areas.

$\omega_{\text{ring}} = 0.23 \kappa; \quad L = f (dM/dt) c^2 T_{\text{grey}}, \quad f \leq 1e-2$ (data-bounded)

GRV-040 (demoted to T2 by GRV-049) / GRV-052, Modeled.

5. THE VACUUM-STIFFNESS TOWER — the v3.17.0 arc

All entries in this section are conditional on FND-037 (Conjecture) at the FND-040 floors.

$$\text{Sigma_eff} = 3.61\text{--}3.70\text{e}35 \text{ J/m}^3 \quad \text{pinned by measurement}$$

FND-030 / ELEC-052/081, Modeled. The historical 5.10e35 registration is demoted: one relation at a dead strand count.

$$\text{Sigma_vac} = \text{kappa_pack} \cdot \text{Sigma_eff}, \quad \text{kappa_pack} \geq 50 \text{ (5\% CS)} / \geq 250 \text{ (continuum)}$$

FND-037/040, Conjecture-grade floor. No ceiling (FND-042: the 1–100 window is HALF-WINDOW — lower edge physics, upper edge grammar).

$$a = (3 K_{\text{me}} / \text{Sigma_vac})^{(1/3)}, \quad T_0 = K_{\text{me}} / a, \quad K_{\text{me}} = T_0 a = m_e c^2 / L = 2.6065\text{e-}14 \text{ J}$$

FND-038/041, Modeled. The m_e -pinned solve; the single spent calibration, re-used never re-fit. $(a, T_0) = (1.63\text{e-}17 \text{ m}, 1599 \text{ J/m}) / (9.53\text{e-}18 \text{ m}, 2734 \text{ J/m})$.

$$\text{delta_D} - \text{delta_f} = -(\text{eps_f}/2)(C_{\text{D}}/C_{\text{f}} - 1), \quad \text{eps_f} = 1/\text{kappa_pack}$$

FND-040, Derived from registered premises (inextensibility + the global tension multiplier). The sign is NEGATIVE: geometric softening.

$$\text{kappa_pack} = (C_{\text{D}}/C_{\text{f}} - 1) / (2 |\text{delta_D} - \text{delta_f}|) \quad \text{THE PIN}$$

FND-046, Modeled. Forward: $\text{delta_A} - \text{delta_F} = -1.25\% \text{ (kappa=50)} / -0.25\% \text{ (kappa=250)}$. Current lattice bound (Bali 2000, $\leq 5\%$) gives $\text{kappa_pack} \geq 12.5$ (FND-047).

$$\text{kappa_pack} = (\ell_q/a) / 43.0 \text{ }^6 \quad \text{the ratio inversion}$$

FND-042, Modeled. Any external determination of ℓ_q/a measures the vacuum packing at $\times 6$ sensitivity.

$$N = C^2 \text{ eta} = 2 g^2 = 3698 \cdot \text{kappa_pack}^{(1/3)} \quad \text{the area-selection composite}$$

FND-043/044, Modeled. The L1 question, the vacuum packing, the amplitude selection and lambda are ONE unknown: what sets g .

6. THE QUANTUM BOUNDARY — what is derived, and what is not

The Schrödinger equation is ADOPTED, not derived; the Born rule is OPEN as a three-part decomposition. What follows is what the corpus does derive.

$$\text{CHSH} \leq 2 \text{ sqrt } 2 \quad \text{the Tsirelson bound, DERIVED}$$

QB-019/020, Derived. Mechanism-unity (one response object per setting per wing) caps CHSH at exactly $2\sqrt{2}$; Gram/TLM verified.

$$P(\text{detect}) = \cos^2(\text{theta}/2) = (1 + a \cdot n)/2 \quad \text{the detection law from strand holonomy}$$

QB-024/025, Derived. The quaternion scalar part IS $\cos(\text{theta}/2)$: the belt trick as the origin of the amplitude.

$$\text{CHSH_local-counting} \leq 2; \quad \text{severed-strand wall } E = -1/3$$

QB-006/013/023, Derived. The counting reading is theorem-capped; $-1/3$ is the signature of severed-strand averaging.

$$S = \pi T A^2 / (2c); \quad A_{\text{hbar}} = \text{sqrt}(2 \text{ hbar } c / (\pi T))$$

HBAR-001, Modeled. L cancels; hbar fixes exactly one number — the amplitude. $A = 2.6348 \cdot \ell_q$ (ELEC-083 identity).

$$\text{rho} = A / \ell_q = 1/\text{sqrt}(2 \pi^2 \alpha) = 2.6348 \Rightarrow 1/\alpha = 2 \pi^2 \text{ rho}^2$$

ELEC-083, Modeled. Under the shared-origin hypothesis α reduces to ONE pure geometric ratio; the hypothesis is carried, not adopted.

7. NUCLEAR STRUCTURE — composition, binding, and the droplet

$M(A, Z)$ = knot count – rope-mode-overlap binding

NUC-001, Derived-structure/Modeled-numeric: masses reproduced to <0.1% given measured binding energies.

atomic masses PREDICTED, C-12 to U-238: 0.00–0.51% (post-NUC-018 correction)

NUC-005 as amended by NUC-018, Modeled. $a_V = 17.77$ with the corrected spacing (2.026 fm) and surface ratio (1.34).

S/V bond ratio = $(d/4)(36 \pi \rho)^{1/3} \cdot \dots$; coordination number z
CANCELS exactly

NUC-015, Derived. Every close-packed lattice gives 1.33–1.34 against an empirical 1.16 — the gap is registered, not hidden.

$n_{\text{cross}} = \rho z d / 4$ bonds cut per unit area, orientation-averaged

NUC-015, Derived.

$R = (3A / 4 \pi \rho)^{1/3}$ droplet radius at saturation density

NUC-022, Modeled.

E_F term: $\langle E \rangle = (3/5) E_F$; symmetry energy from $N \neq Z$ label counting

NUC-010/019/022, Modeled. The $N=Z$ preference emerges from unlike-label bonding.

8. MATTER, ATOMS, AND CHEMISTRY

shell capacities = $2 n^2$ (n^2 rope modes $\times$ two spin states)

CHEM-STRUCT-001, Derived (counting).

$R \sim a \sqrt{N / 4 \pi}$ atomic pattern radius from rope count

FND-MATTER-002, Modeled — a consistency WINDOW, not a prediction (two free inputs, one target). Numerological fixes explicitly rejected.

N fixed by the interpenetrability coverage threshold (a geometric number, not a parameter)

FND-MATTER-004, Modeled.

$\lambda = \pi (r/a)^2 = 2.776e-6$ zero-point suppression, dilute-perturbation Casimir

FND-MATTER-056, Modeled — a QUANTIFIED NULL: $4.16\times$ from target, with the $O(1)$ rescue pre-refused. λ REMAINS OPEN.

IRREDUCIBILITY: the absolute mesh scale a is NOT derivable from the postulates.

FND-MATTER-005, Derived (a theorem about the framework). One calibration is spent; the corpus honours this rather than quietly contradicting it.

9. THERMODYNAMICS AND STATISTICAL MECHANICS

$Z = \prod \sqrt{2 \pi T / (K \lambda_{k,k})}$ Gaussian partition function

THM-002, Derived; free energy, entropy and energy follow.

$C = 1/2$ per mode 2D Dulong-Petit

THM-003, Derived.

$$T_{\text{BKT}} = \pi K / 2; \quad \eta(T_{\text{BKT}}) = 1/4; \quad \text{jump} = 2 T_{\text{BKT}} / \pi$$

THM-004/005, Derived. Standard BKT phenomenology on the corpus's own stiffness.

10. THE FALSIFIABLE RELATIONS (what could kill the programme)

$d \ln \alpha / d \ln G$ = fixed scale-free ratio (-1, -2, or -3 by l_q composition)

PRED-003, Modeled — the corpus's sole T1. A measured nonzero ratio IDENTIFIES the drift channel and, through the FND-042 inversion, reads κ_{pack} (FND-043).

$|G\dot{G}| > 1.6e-18$ /yr at 3 sigma REFUTES (decision window 2027–2030, J1713)

PRED-003-CONF/META/J1713, Modeled.

$\Delta_A - \Delta_F = -1.25\% / -0.25\%$ long-distance Casimir-scaling violation, SU(3)

FND-046/047, Modeled. Kill conditions: quartic-DOMINANT violation kills; POSITIVE deviation kills the derived sign; $\epsilon_f > 5\%$ kills the floor.

$\sigma_k < k \sigma_1$ bundles bind (k unit windings attract)

FND-048, Modeled. The qualitative bundle prediction; matches every dataset under both candidate laws. The (N,k) magnitude is the named next frontier.

$\Delta n < 0$ with 3:1 anisotropy vacuum birefringence (PVLAS-class)

EM-RECON-016, Derived given the mapping; magnitude Sigma-suppressed.

11. THE OPEN NUMBERS — what the corpus does not have

An honest compendium ends with the blanks. Four Conjectures, six Open claims, thirty-four kept failures stand in the registry; the formula-level gaps are these:

$g = l_q/a$ — the mesoscopic source length. UNEXPLAINED.

FND-044. Five former questions collapse to it. Its best-shaped mechanism (defect-log energy budget) is EXCLUDED structurally: FND-045, Failed-and-kept, demand $2.2\text{--}3.2 m_e c^2$ blind.

λ — the zero-point suppression. OPEN.

FND-MATTER-050/055/056. Reframed: not “what weights the term” but “what suppresses mesh-scale vacuum energy by $\sim 1e-5$ ”; — the matter sector's own hierarchy problem.

$\hbar$ — not derived. The question has a SHAPE: what selects the area $l_q^2/(4\pi\alpha)$?

GRV-093 / FND-043. Narrowed to the locking normalization N; not closed.

m_e , and the absolute scale — irreducible by theorem, one calibration spent.

FND-MATTER-005 / FND-017.

Born rule, Schrödinger equation — adopted/open, not derived.

QB-007/008; see KNOWN_LIMITATIONS.md, which is the document to read before this one.

Compiled 2026-08-11 at v3.17.0. Provenance for every line is the claim ID given beneath it; run `tools/verify_corpus.py` and the cited benchmarks to check any of them. Failures are kept on display by design — see `KNOWN_LIMITATIONS.md` and the Failed-and-kept claims in `claims.yaml`.

Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, `rope_solver`, file names, claim IDs) are unchanged.